MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 18: Dynamic Treatment Effects & Long-Term Effects

Vasilis Syrgkanis

Stanford University

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Part I: Introduction & Motivation

From Static to Sequential Interventions

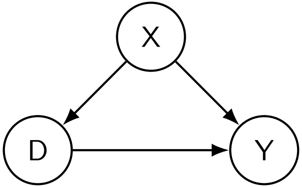
Goals of this Lecture

Today we move from single-shot interventions to sequences of treatments over time.

1. **Dynamic Treatment Regimes (DTRs):** What are they and why are they a crucial tool for real-world decision-making?
2. **Identification & Estimation:** How can we identify and estimate the mean counterfactual outcome under a specific DTR using the sequential g-formula and debiased machine learning?
3. **Continuous Treatments:** How can we handle continuous treatments in a dynamic setting using a partialling-out approach (Dynamic DML)?
4. **Surrogates:** How can we use short-term surrogate outcomes to estimate long-term treatment effects, and what are the pitfalls?

From Static to Dynamic Treatments

So far, we have mostly focused on units being treated at **one point in time**, after which we observe an outcome.



But in many settings, units are treated **multiple times over time**. The treatment at one stage can depend on the history of treatments and outcomes up to that point. This is a **Dynamic Treatment Regime**.

Motivating Example 1: Healthcare

Scenario: A cancer patient might receive multiple treatments sequentially (e.g., first-line therapy, then second-line therapy based on response).

Application: Cancer Treatment Regimes (e.g., Talbot, Moodie, & Diorio, 2023)

Cancer patients typically undergo multiple lines of therapy (e.g., chemotherapy, surgery, immunotherapy, radiation therapy, hormonal therapy). These treatments are typically administered in sequence based on prior treamtents and on observed response factors / adverse events.

Causal Question: What would be the mean counterfactual outcome if all patients first received immunotherapy and then chemotherapy?

Recent Example: DTRs in Mental Health

Question: What is the optimal sequence of treatments for patients with Major Depressive Disorder (MDD) who do not respond to an initial antidepressant?

- **Domain:** Psychiatry / Depression Treatment
- **Data Source:** The Sequenced Treatment Alternatives to Relieve Depression (STAR*D) trial—the largest multi-step clinical trial for depression ($n = 4{,}041$).
- **Analysis:** Szmulewicz et al. (2023) emulated a target trial using STAR*D data to compare different dynamic treatment strategies.

STAR*D: Dynamic Treatment Strategies Compared

Goal. Evaluate the counterfactual outcome for three treatment strategies for 971 patients failing initial citalopram:

1. **Sequential Monotherapy:** Switch to a different single antidepressant.
2. **Sequential Dual Therapy (SD):** Switch to a combination of two non-SSRI drugs.
3. **Guidelines-Based Strategy (DTR):** Adapts to intermediate response:
   - *Partial response* $\rightarrow$ Augment current drug with a second medication.
   - *No response / intolerance* $\rightarrow$ Switch to a new antidepressant.

STAR*D: Key Results

Strategy	**9-Month Remission**	**RR vs. Monotherapy**
Sequential Monotherapy	43.5%	1.00 (ref)
Sequential Dual Therapy	47.6%	1.09 (0.90–1.38)
Guidelines-Based DTR	**53.2%**	**1.22 (0.96–1.46)**

Takeaway

The **guidelines-based DTR** achieved the highest remission rate and the lowest risk of serious adverse events (RR $= 0.62$). Formalizing clinical guidelines as a DTR and evaluating them with causal methods provided actionable evidence for sequential depression management.

Motivating Example 2: Marketing

Scenario: A customer's journey on a website involves multiple touchpoints. We can show them different ads or offers at different times.

Application: Customer Lifetime Value

Firms want to optimize a sequence of marketing actions (e.g., ads, discounts) to maximize a customer's long-term value. The decision at each stage depends on the customer's browsing history and prior responses.

Causal Question: What is the expected 2-year revenue from a customer if we show them a banner ad on their first visit, and then a 10% discount if they add an item to their cart?

Motivating Example 3: B2B Operations

Scenario: A large cloud provider (like Microsoft Azure) wants to help enterprise customers adopt its services.

Application: Azure Customer Growth (e.g., Battocchi et al., 2021)

Microsoft offers free credits to new customers and can assign a sales specialist to help them. These interventions happen at different stages of the customer lifecycle.

Causal Question: What is the expected customer spend if we offer \$10k in initial credits, and then assign a sales specialist if their usage exceeds a certain threshold in the first 6 months?

Motivating Example 4: Labor Economics

Scenario: A government wants to design a program to help unemployed individuals find jobs.

Real-World Application: Job Training Programs (Bodory et al., 2022)

An individual might first be offered a resume-building workshop. If they are still unemployed after 3 months, they might be offered a more intensive vocational training program.

Causal Question: What is the average employment rate after one year if we follow this sequential training strategy?

Poll 1: Your Experience

Have you ever worked on a problem that involved sequential decision-making or dynamic treatments?

A. Yes, frequently.	B. Yes, once or twice.
C. No, but I see the relevance.	D. Not sure.

Part II: A Mathematical Abstraction

A Two-Period Model of Sequential Treatment

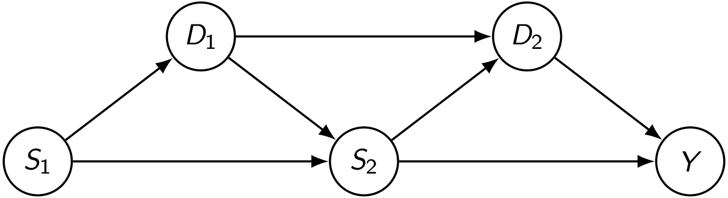
A Two-Period Model

We can abstract all of these scenarios with the following mathematical notation for a two-period study:

- S_1: The initial state (pre-treatment covariates) of a unit.
- D_1: The first period treatment, which may be chosen based on S_1.
- S_2: The next period's state. This is affected by both S_1 and D_1. It can contain qualitatively different variables than S_1. It can even contain S_1, D_1 as coordinates.
- D_2: The second period treatment, which may be chosen based on the state S_2.
- Y: The final outcome of interest, which is affected by S_2 and D_2.

The Causal DAG for a Two-Period Model

The causal relationships between variables in a two-period study:



Note: S_2 is both a **mediator** (on the path $D_1 \rightarrow S_2 \rightarrow Y$) and a **confounder** (common cause of D_2 and Y). This dual role is the key challenge.

The Goal: Estimating Joint Interventions

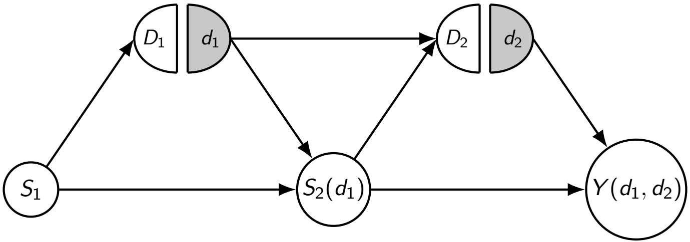
Our goal is to estimate the mean potential outcome under a **fixed** joint intervention on the treatment sequence.

Estimand: $\theta = \mathbb{E}[Y(d_1, d_2)]$

This represents the expected outcome if we were to set the first treatment to d_1 for everyone, and the second treatment to d_2 for everyone.

The SWIG for a Two-Period Intervention

To reason about the intervention $\text{fix}(D_1 = d_1, D_2 = d_2)$, we draw the Single World Intervention Graph (SWIG).



Revisiting Examples: Healthcare (STAR*D)

Let's map the STAR*D trial to our abstraction:

- S_1: Patient baseline characteristics (age, gender, depression severity).
- D_1: First-line therapy (e.g., immunotherapy, d_1).
- S_2: Patient status after 3 months (e.g., symptom remission, side effects).
- D_2: Second-line therapy if no remission (e.g., chemotherapy, d_2).
- Y: Patient outcome at 1 year (e.g., depression-free survival).

Goal: Estimate $\mathbb{E}[Y(\text{immuno}, \text{chemo})]$.

Revisiting Examples: Marketing (LTV)

Mapping the customer lifetime value problem:

- S_1: Customer's initial state (demographics, referral source).
- D_1: First marketing action (e.g., banner ad, d_1).
- S_2: Customer's state after first action (e.g., pages visited, items in cart).
- D_2: Second marketing action (e.g., 10% discount, d_2).
- Y: Customer's 2-year revenue.

Goal: Estimate $\mathbb{E}[Y(\text{banner}, \text{discount})]$.

Part III: The Identification Challenge

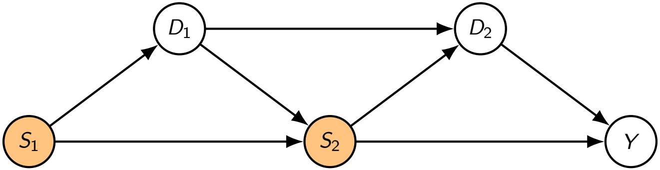
Why Standard Methods Fail

Why Not Just Condition on All States?

A naive approach: condition on the full history of states (S_1, S_2).

$$\mathbb{E}[Y|D_1 = d_1, D_2 = d_2, S_1, S_2] \quad \neq \quad \mathbb{E}[Y(d_1, d_2)|S_1, S_2]$$

This is incorrect! Conditioning on S_2 (a post-treatment variable) blocks the effect of D_1 that is mediated through S_2. It induces collider bias.

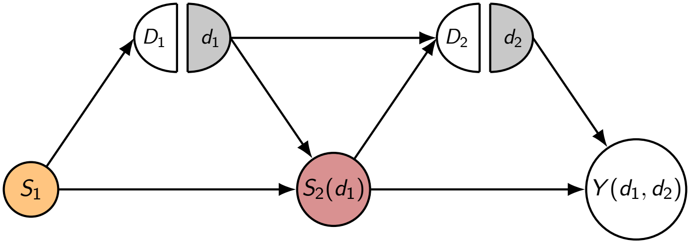


Why Not Just Condition on All States?

This is incorrect! Conditional exogeneity cannot be checked on the SWIG:

$$Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1, S_2$$

The variable S_2 does not even appear on the SWIG; only $S_2(d_1)$!



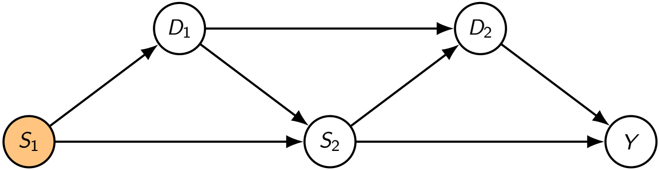
Mantra: If you cannot verify it on the SWIG don't use it for identification.

Why Not Just Condition on Pre-Treatment State?

What if we only condition on the baseline state S_1?

$$\mathbb{E}[Y|D_1 = d_1, D_2 = d_2, S_1] \quad \neq \quad \mathbb{E}[Y(d_1, d_2)|S_1]$$

This is also incorrect! S_2 is a common cause of D_2 and Y. Since we haven't conditioned on S_2, there is an unblocked backdoor path from D_2 to Y through S_2, creating confounding.

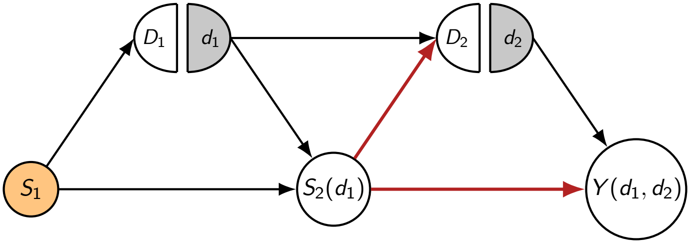


Why Not Just Condition on Pre-Treatment State?

This is incorrect! Conditional exogeneity cannot be verified on the SWIG:

$$Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1$$

d-separation does not hold; there is an open path $D_2 \leftarrow S_2(d_1) \rightarrow Y(d_1, d_2)$!



Poll 2: The Main Challenge

What do you think is the fundamental challenge in identifying dynamic treatment effects?

A. Lack of randomization.

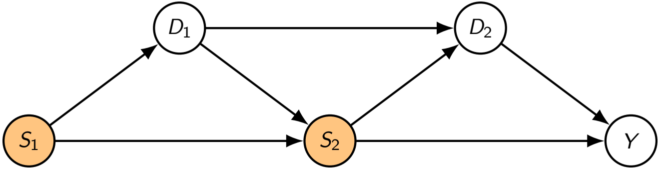
B. Time-varying confounding.

C. Variables that are both mediators and confounders.

D. All of the above.

What Effects Can we Identify by Conditioning?

The effect on which variable V_1 on what variable V_2 can we identify by conditioning?



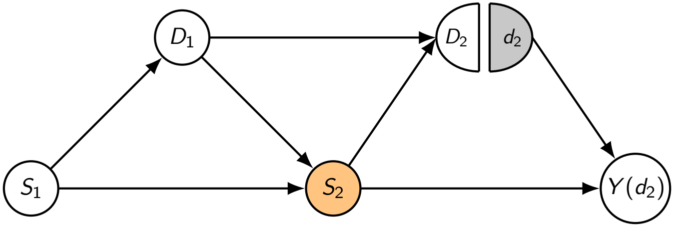
Part IV: Identification via Backwards Recursion

The Sequential G-Formula

Effects of Just Intervening on Last Period

The effect on only intervening on D_2 on Y can be identified by conditioning on S_2.

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2$$

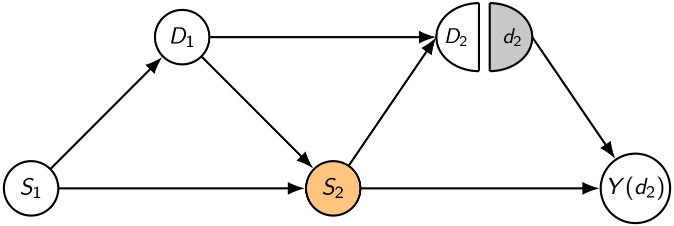


Conditional mean counterfactual outcomes can be identified by conditioning on S_2:

$$\underbrace{Q_2(S_2, d_2) := \mathbb{E}[Y(d_2) \mid S_2]}_{\substack{\text{Forecasted counterfactual outcome based on } S_2 \\ \text{under second period intervention}}} \quad = \quad \underbrace{\mathbb{E}[Y \mid S_2, D_2 = d_2]}_{\substack{\text{Forecasted counterfactual outcome based on } S_2 \\ \text{for samples treated with } D_2 = d_2 \text{ in second period}}}$$

Effect of D_1 on $Y(d_2)$ is fully mediated by S_2

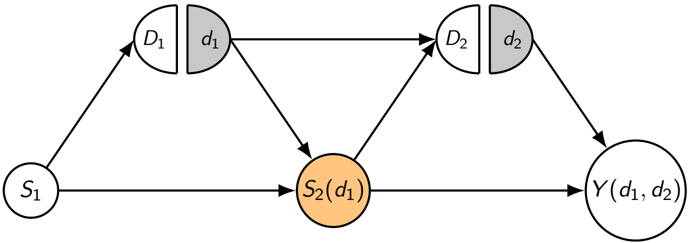
Assuming we are in a world where we intervene in the second stage.
The effect of D_1 on $Y(d_2)$ is fully mediated by S_2.



The effect of D_1 on $Y(d_2)$, is equal to the effect of D_1 on the "forecasted" counterfactual outcome, forecasted from S_2, i.e. $Q_2(S_2, d_2)$:

$$\underbrace{Q_1(S_1, d_1) := \mathbb{E}[Y(d_1, d_2) \mid S_1]}_{\text{Effect of fix}(D_1 = d_1) \text{ on } Y(d_2)} = \underbrace{\mathbb{E}[Q_2(S_2(d_1), d_2) \mid S_1]}_{\substack{\text{Effect of fix}(D_1 = d_1) \text{ on forecasted } Y(d_2) \\ \text{forecasted from } S_2}}$$

Formally!

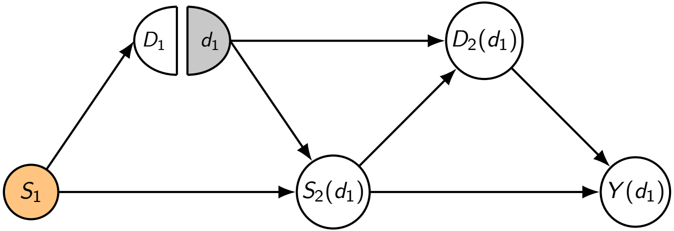


$$
\begin{aligned}
\mathbb{E}[Y(d_1, d_2) \mid S_1] &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1)] \mid S_1] && \text{(tower rule)} \\
&= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && (d\text{-separation in SWIG}) \\
&= \mathbb{E}[\mathbb{E}[Y(D_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\
&= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\
&= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1)] \mid S_1] \\
& && (d\text{-separation in SWIG from prev. slide})
\end{aligned}
$$

Effects of Just Intervening on First Period

The effect of only intervening on D_1 on S_2 can be identified by conditioning on S_1.

$$S_2(d_1) \perp\!\!\!\perp D_1 \mid S_1$$



The same holds for any function of S_2, e.g., $Q_2(S_2, d_2)$:

$$Q_2(S_2(d_1), d_2) \perp\!\!\!\perp D_1 \mid S_1$$

Thus we can identify conditional mean counterfactual outcomes by conditioning:

$$Q_1(S_1, d_1) = \mathbb{E}[Y(d_1, d_2) \mid S_1] = \mathbb{E}[Q_2(S_2(d_1), d_2) \mid S_1] = \mathbb{E}[Q_2(S_2, d_2) \mid S_1, D_1 = d_1]$$

Putting it All Together: Identification via Backwards Recursion

1. Build forecasting model of outcome from second period (state, treatment):

$$Q_2(S_2, D_2) = \mathbb{E}[Y \mid S_2, D_2]$$

2. Construct adjusted outcome for each sample based on S_2: outcome we expect for the unit if we intervene in second period and fix$(D_2 = d_2)$

$$Y_{adj} = Q_2(S_2, d_2)$$

3. Build forecasting model of **adjusted outcome** from first period (state,treatment):

$$Q_1(S_1, D_1) = \mathbb{E}[Y_{adj} \mid S_1, D_1]$$

4. Construct adjusted outcome for each sample based on S_1: outcome we expect for the unit if we **also intervene** in first period and fix$(D_1 = d_1)$

$$Y_{adj} = Q_1(S_1, d_1)$$

5. Average the final adjusted outcomes:

$$\theta = \mathbb{E}[Y(d_1, d_2)] = \mathbb{E}[Y_{adj}]$$

The Sequential G-Formula (Robins, 1986)

This backwards recursion leads to the celebrated **sequential g-formula**

$$\theta := \mathbb{E}[Y(d_1, d_2)] = \mathbb{E}\Big[Q_1(S_1, d_1)]$$

where Q_1 is defined as a sequence of nested regression models:

$$Q_2(S_2, D_2) = \mathbb{E}[Y|S_2, D_2]$$
$$Q_1(S_1, D_1) = \mathbb{E}[Q_2(S_2, d_2) \mid S_1, D_1]$$

Reminder: one period g-formula:

$$\theta = \mathbb{E}[Y(d_1)] = \mathbb{E}[g(S_1, d_1)] \qquad g(S_1, D_1) = \mathbb{E}[Y \mid S_1, D_1]$$

Part V: Estimation

From G-Computation to Doubly Robust DTR

G-Estimation Algorithm

The G-formula gives us a simple plug-in estimation algorithm:

1. Train a model $\hat{Q}_2$ to predict Y from (S_2, D_2).
2. Predict $\hat{Y}_{2,i} = \hat{Q}_2(S_{2,i}, d_2)$ for all samples i.
3. Train a model $\hat{Q}_1$ to predict these $\hat{Y}_2$ values from S_1 and D_1.
4. Predict $\hat{Y}_{1,i} = \hat{Q}_1(S_{1,i}, d_1)$ for all samples.
5. Estimate θ by averaging the final predictions: $\hat{\theta} = \frac{1}{n} \sum_i \hat{Y}_{1,i}$.

Problem: If we use flexible Machine Learning models for $\hat{Q}_1, \hat{Q}_2$, the plug-in estimator $\hat{\theta}$ will be heavily biased and we can't get valid confidence intervals.

Debiasing Step 1: Correcting for Q_1

The plug-in moment is sensitive to errors in Q_1

$$\theta = \mathbb{E}[Q_1(S_1, d_1)]$$

Add a correction to make it insensitive to Q_1, which is a regression function:

$$Q_1(S_1, D_1) := \mathbb{E}[Q_2(S_2, d_2)|S_1, D_1]$$

Step 1: Debiasing Q_1 (Doubly Robust Formula)

$$\underbrace{Q_1(S_1, d_1)}_{\text{Original formula}} + \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)}}_{\text{Inverse propensity}} \cdot \underbrace{\left(Q_2(S_2, d_2) - Q_1(S_1, D_1)\right)}_{\text{Residual of regression defining } Q_1}$$

Problem: We are still left with the nuisance Q_2 inside the correction, which we have **not yet debiased**. Formula is still sensitive to Q_2.

Debiasing Step 2: Correcting for Q_2

We now add a second correction term to debias Q_2:

Step 2: Debiasing f_2 (Final Doubly Robust Formula)

$$
\begin{aligned}
Y_{DR} = & Q_1(S_1, d_1) + \frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)} \left(Q_2(S_2, d_2) - Q_1(S_1, D_1)\right) \\
& + \underbrace{\frac{\mathbb{I}(D_2 = d_2)}{\Pr(D_2 = d_2 \mid S_2)}}_{\text{New IPS for period 2}} \cdot \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 \mid S_1)}}_{\text{IPS already multiplying } Q_2} \cdot \underbrace{\left(Y - Q_2(S_2, D_2)\right)}_{\text{Residual of regression defining } Q_2}
\end{aligned}
$$

Multiply Robust Formula

The Multiply Robust estimator is defined based on the formula

$$\theta = \mathbb{E}[Y_{DR}] \tag{1}$$

- We also need to estimate propensity functions p_1, p_2 via ML classification:

$$p_t(S_t) = \Pr(D_t = d_t \mid S_t)$$

- This formula is insensitive to all nuisance functions (regressions and propensities)

We estimate all these functions in a cross-fitting manner, calculate estimated doubly robust pseudo-outcome $\hat{Y}_{DR}$ and calculate final estimate

$$\hat{\theta}_{DR} = \mathbb{E}_n[\hat{Y}_{DR}]$$

Dynamic Doubly Robust Estimator: Pseudocode

```python
import numpy as np
from sklearn.model_selection import KFold
# Nuisance models: f2, f1 (regressors), e1, e2 (classifiers)
psi_values = np.zeros(n)
kf = KFold(n_splits=K)
for train_idx, test_idx in kf.split(S1):
    # Stage 2: fit f2(S2, D2) -> Y, and propensity p2(S2) -> D2
    f2_model.fit(np.c_[S2[train_idx], D2[train_idx]], Y[train_idx])
    p2_model.fit(S2[train_idx], D2[train_idx])
    # Stage 1: fit f1(S1, D1) -> f2(S2, d2) and propensity p1(S1) -> D1
    f2_pseudo = f2_model.predict(np.c_[S2[train_idx], np.full(len(train_idx), d2)])
    f1_model.fit(np.c_[S1[train_idx], D1[train_idx]], f2_pseudo)
    p1_model.fit(S1[train_idx], D1[train_idx])

    # Predict on test fold
    f2_hat = f2_model.predict(np.c_[S2[test_idx], D2[test_idx]])
    f1_hat = f1_model.predict(np.c_[S1[test_idx], D1[test_idx]])
    f2_hat_d2 = f2_model.predict(np.c_[S2[test_idx], np.full(len(test_idx), d2)])
    f1_hat_d1 = f1_model.predict(np.c_[S1[test_idx], np.full(len(test_idx), d1)])
    p1_hat = p1_model.predict_proba(S1[test_idx].reshape(-1,1))[:, d1]
    p2_hat = p2_model.predict_proba(np.c_[S1[test_idx], D1[test_idx], S2[test_idx]])[:, d2]

    # Doubly robust score
    w1 = (D1[test_idx]==d1) / np.clip(p1_hat, 0.01, 1)
    w2 = (D2[test_idx]==d2) / np.clip(p2_hat, 0.01, 1)
    psi_values[test_idx] = f1_hat_d1 + w1 * (f2_hat_d2 - f1_hat) + w2 * w1 * (Y[test_idx] - f2_hat)

theta_hat = np.mean(psi_values)
se_hat = np.std(psi_values) / np.sqrt(n)
ci = (theta_hat - 1.96 * se_hat, theta_hat + 1.96 * se_hat)
```

Asymptotic Normality Theorem

Main Theorem (Dynamic DR asymptotic normality)

Under mild regularity conditions, sequential conditional ignorability

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2 \qquad S_2(d_2) \perp\!\!\!\perp D_1 \mid S_1$$

and **strict** sequential overlap, i.e., $\Pr(D_t \mid S_t) \in (\epsilon, 1-\epsilon)$, if the Q functions and the propensities are consisten $\mathrm{RMSE}(\hat{Q}_t), \mathrm{RMSE}(\hat{p}_t) \to 0$ and

$$\sqrt{n}\left(\mathrm{RMSE}(\hat{Q}_1)\,\mathrm{RMSE}(\hat{p}_1) + \mathrm{RMSE}(\hat{Q}_2)\,\mathrm{RMSE}(\hat{p}_1) + \mathrm{RMSE}(\hat{Q}_2)\,\mathrm{RMSE}(\hat{p}_2)\right) \to 0, \tag{Product Rate Condition}$$

then the Dynamic Doubly Robust Estimator with cross-fitting satisfies

$$\sqrt{n}(\hat{\theta}_{\mathrm{DR}} - \mathbb{E}[Y(d_1, d_2)]) \Rightarrow \mathcal{N}(0, \mathrm{Var}(Y_{DR})).$$

Standard errors can be estimated as: $\widehat{\mathrm{se}} = \sqrt{\mathrm{Var}_n(\hat{Y}_{DR})/n}$, and 95% CI $= \hat{\theta}_{\mathrm{DR}} \pm 1.96\widehat{\mathrm{se}}$.

Part VI: Evaluating Adaptive Policies

Counterfactual Treatments Can Depend on States

What if the Policy is Adaptive?

So far, we considered a fixed policy (d_1, d_2). What if the policy is **adaptive** (or dynamic), where the treatment decision depends on the current state?

An adaptive policy π is a set of decision rules:

$$\pi = (\pi_1, \pi_2) \quad \text{where} \quad d_1 = \pi_1(S_1) \text{ and } d_2 = \pi_2(S_2)$$

Example: In the STAR*D trial, a policy might be:

- $\pi_1(S_1)$: Always start with Citalopram.
- $\pi_2(S_2)$: If patient has not remitted (S_2), switch to Bupropion; otherwise, continue Citalopram.

Our goal is now to estimate the mean outcome under this adaptive policy:

$$\theta^\pi = \mathbb{E}[Y(\pi_1(S_1), \pi_2(S_2(d_1)))].$$

Estimation for Adaptive Policies

The G-formula and the doubly robust approach extend naturally.

G-Computation: Replace fixed d_t with the policy function $\pi_t(S_t)$ at each step.

$$Q_2(S_2, D_2) = \mathbb{E}[Y \mid S_2, D_2]$$
$$Q_1(S_1, D_1) = \mathbb{E}[Q_2(S_2, \pi_2(S_2))|S_1, D_1]$$

Doubly Robust Formula for Adaptive Policy π

$$\begin{aligned} Y_{DR} = & Q_1(S_1, \pi_1(S_1))) \\ & + \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)}(Q_2(S_2, \pi_2(S_2)) - Q_1(S_1, D_2)) \\ & + \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} \frac{\mathbb{I}(D_2 = \pi_2(S_2))}{\Pr(D_2 = \pi_2(S_2) \mid S_2)}(Y - Q_2(S_2, D_2)) \end{aligned}$$

Generalization to $T > 2$ Periods

The entire framework of backwards recursion, G-computation, and doubly robust estimation generalizes to any number of time periods T.

G-Formula Recursion:

$$
\begin{aligned}
Q_T(S_T, D_T) &= \mathbb{E}[Y | S_T, D_T] \\
Q_t(S_t, D_t) &= \mathbb{E}[Q_{t+1}(S_{t+1}, \pi_{t+1}(S_{t+1})) | S_t, D_t] \quad \text{for } t = 1, ..., T-1 \\
\theta &= \mathbb{E}[Q_1(S_1, \pi_1(S_1))]
\end{aligned}
$$

Doubly Robust Moment: The moment function will have $T+1$ terms, with each term correcting for the estimation error of a nuisance function Q_t from period $t = 1, ..., T$.

Part VII:Continuous Treatments

Dynamic Double Machine Learning

What if Treatments are Continuous?

The inverse propensity weighting approach fails for continuous (or high-dimensional) treatments, as the probability of observing the exact treatment value d_t is zero.

What is the analogue of the "Residual-on-Residual" or "Partialling-Out" (FWL) approach for the dynamic treatment regime?

This leads to an alternative identification and estimation strategy based on so-called **"blip" effects**.

Identification via "Blip" Effects

Instead of modeling the conditional outcome, we can model the incremental **effect** of each treatment stage.

Step 2 (Blip): Define the effect of the second treatment as:

$$\alpha_2(S_2, d_2) := \mathbb{E}[Y(D_1, d_2) - Y(D_1, 0)|S_2]$$

This is the "blip" effect of D_2. We can identify it by solving a standard single-treatment causal problem, conditioning on S_2.

Adjusted Outcome: We can then create an adjusted outcome that removes the effect of the observed treatment D_2 and adds back the effect of the target treatment d_2:

$$Y^{\text{adj}} := Y - \alpha_2(S_2, D_2) + \alpha_2(S_2, d_2)$$

Identification via "Blip" Effects

Step 1 (Blip): Define the effect of the first treatment as:

$$\alpha_1(S_1, d_1) := \mathbb{E}[Y(d_1, d_2) - Y(0, d_2)|S_1]$$

This is the "blip" effect of D_1. We can identify it by solving a standard single-treatment causal problem, conditioning on S_1 on the adjusted outcome from the previous period.

Adjusted Outcome: We can then create an adjusted outcome that removes the effect of the observed treatment D_1 and adds back the effect of the target treatment d_1:

$$Y^{\text{adj}} := Y^{\text{adj}} - \alpha_1(S_1, D_1) + \alpha_1(S_1, d_1)$$

Dynamic DML Algorithm (Lewis, Syrgkanis (2021))

Assuming linear blip effects:

$$\alpha_t(S_t, d_t) = \delta_t d_t.$$

is the sequential analogue of the Partially Linear Model.

Dynamic DML (2-Stage, Linear Blips)

1. **Stage 2**: Residualize $\tilde{Y}_2 = Y - \hat{\mathbb{E}}[Y|S_2]$, $\tilde{D}_2 = D_2 - \hat{\mathbb{E}}[D_2|S_2]$. Regress $\tilde{Y}_2$ on $\tilde{D}_2$ to get $\hat{\delta}_2$.
2. Create adjusted outcome: $Y^{\text{adj}} = Y - \hat{\delta}_2 D_2 + \hat{\delta}_2 d_2$.
3. **Stage 1**: Residualize $\tilde{Y}_1 = Y^{\text{adj}} - \hat{\mathbb{E}}[Y^{\text{adj}}|S_1]$, $\tilde{D}_1 = D_1 - \hat{\mathbb{E}}[D_1|S_1]$. Regress $\tilde{Y}_1$ on $\tilde{D}_1$ to get $\hat{\delta}_1$.
4. Create adjusted outcome: $Y^{\text{adj}} = Y^{\text{adj}} - \hat{\delta}_1 D_1 + \hat{\delta}_1 d_1$.
5. Average the final adjusted outcomes: $\hat{\tau} = \mathbb{E}_n[Y^{\text{adj}}]$

Dynamic DML Algorithm as an Insentive Formula

We can view this algorithm as solving the empirical version of the normal equations of the corresponding OLS problems:

$$\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2)\,\tilde{D}_2] = 0$$
$$\mathbb{E}[(\tilde{Y}_1 - \delta_1 \tilde{D}_1)\,\tilde{D}_1] = 0$$

The target quantity $\tau = \mathbb{E}[Y(d_1, d_2)]$ is given as a function of δ_1, δ_2:

$$\tau = \mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1]$$

Dynamic DML Algorithm as an Insentive Formula

Note that $\tilde{Y}_1$ implicitly depends on δ_2. We make it explicit:

$$\tilde{Y}_1 = \tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)}, \qquad \tilde{V}^{(1)} = V - \mathbb{E}[V \mid S_1]$$

Which gives the system of equations:

$$\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2)\, \tilde{D}_2] = 0$$
$$\mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1)\, \tilde{D}_1] = 0$$

Both of these equations are insensitive to the nuisance functions, which correspond to the regression problems: $\mathbb{E}[Y \mid S_2]$, $\mathbb{E}[Y \mid S_1]$, $\mathbb{E}[D_2 \mid S_2]$, $\mathbb{E}[D_2 \mid S_1]$, $\mathbb{E}[D_1 \mid S_1]$.

Dynamic DML Algorithm as an Insentive Formula

Parameters $\theta, \delta_1, \delta_2$ are solution to the following equations, which are insensitive to nuisance functions

$$
\begin{aligned}
\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2)\,\tilde{D}_2] &= 0 \\
\mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1)\,\tilde{D}_1] &= 0 \\
\mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau] &= 0
\end{aligned}
$$

The estimated parameters $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$ based on the dynamic DML algorithm are the solution to the empirical analogue of these equations.

Generic Moment Equation Based Estimator (Chernozhukov et al, 2018)

Suppose that a vector of parameters θ_0 is the solution to a vector of equations:

$$\theta_0 \text{ is solution to } \mathbb{E}[m(Z;\theta,g_0)] = 0$$

for some known *moment functions* m that depend on nuisance functions g_0 but are insensitive to the nuisance functions.

Consider the estimated parameters $\hat{\theta}$ that are the solution to the empirical plug-in analogue of these equations:

$$\hat{\theta} \text{ is solution to } \mathbb{E}_n[m(Z;\theta,\hat{g})] = 0$$

where nuisances $\hat{g}$ are estimated in a cross-fitting manner.

Generic Asymptotic Normality (Chernozhukov et al, 2018)

Under regularity conditions and sufficiently fast RMSE rates ($\ll n^{-1/4}$) on the nuisances, the parameters $\hat{\theta}$ are jointly asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta) \to N(0, J^{-1}\Sigma J^{-\top})$$

where J is the Jacobian of the equations with respect to the parameters:

$$J = \nabla_\theta \mathbb{E}[m(Z; \theta_0, g_0)]$$

and Σ is the covariance matrix of the equations:

$$\Sigma = \mathbb{E}[m(Z; \theta_0, g_0) m(Z; \theta_0, g_0)']$$

Standard errors can be calculated as $\sqrt{\hat{V}/n}$ using estimated variance $\hat{V} = \hat{J}^{-1}\hat{\Sigma}\hat{J}^{-\top}$:

$$\hat{J} = \nabla_\theta \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g})] \qquad \hat{\Sigma} = \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g}) m(Z; \hat{\theta}, \hat{g})']$$

Confidence Intervals for Dynamic DML Algorithm

Parameters $\theta = (\tau, \delta_1, \delta_2)$ are solution to the following equations, which are insensitive to nuisance functions

$$\mathbb{E}[m(Z;\theta,g_0)] = 0 \qquad m(Z;\theta,g_0) = \begin{pmatrix} (\tilde{Y}_2 - \delta_2 \tilde{D}_2)\, \tilde{D}_2 \\ (\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1)\, \tilde{D}_1 \\ Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau \end{pmatrix}$$

g_0 are the regression functions: $\mathbb{E}[Y \mid S_2]$, $\mathbb{E}[Y \mid S_1]$, $\mathbb{E}[D_2 \mid S_2]$, $\mathbb{E}[D_2 \mid S_1]$, $\mathbb{E}[D_1 \mid S_1]$

The estimated parameters $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$ based on the dynamic DML algorithm are the solution to the empirical analogue of these equations for all parameters.

We can apply the general framework to these equations to derive confidence intervals and standard errors.

Takeaways

- Dynamic Treatment Regimes are a powerful framework for modeling and optimizing sequential decisions in many real-world domains.
- Simple conditioning fails due to time-varying confounding, where intermediate states are both mediators and confounders.
- The **sequential G-formula** provides a general identification strategy via backwards recursion.
- We can construct **doubly robust estimators** for DTRs that allow for flexible ML nuisance models while retaining valid inference.
- For continuous treatments, **Dynamic DML** provides an alternative estimation strategy based on partialling-out blip effects.

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